

PROGRAM - 06

Deploy an automated build pipeline using Docker hub

Structure of the program:

- Dockerfile
- Package.json
- Package-lock.json
- Node_modules

Steps:

1. Create the personal access token in github

- Create a repository(i.e program-6)
- Navigate to the settings -> Developers settings -> Generate tokens
- Generate the token with specific scope:
 - Write:packages
 - Read:packages

2. Create folder program-6 and write the appropriate files required

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~$ mkdir program-6
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~$ cd program-6
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program-6$ nano Dockerfile
```

3. Create a Docker file and build the images

```
FROM node:18-alpine
WORKDIR /app
COPY package*.json .
RUN npm ci --only=production
COPY . .
```

```

_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program-6$ docker build -t program-6 .
[+] Building 9.7s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 130B                               0.0s
=> [internal] load metadata for docker.io/library/node:18-alpine  0.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                     0.0s
=> [internal] load build context                                  0.2s
=> => transferring context: 2.27MB                                 0.2s
=> [1/5] FROM docker.io/library/node:18-alpine                   0.0s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> [3/5] COPY package*.json .                                     0.1s
=> [4/5] RUN npm ci --only=production                             8.6s
=> [5/5] COPY . .                                                  0.5s
=> exporting to image                                              0.2s
=> => exporting layers                                             0.2s
=> => writing image sha256:94d14cb794cf4bc233f72a18af59d980ca6ca87845572 0.0s
=> => naming to docker.io/library/program-6                      0.0s

```

4. Tag the image for docker hub and login to the docker hub using PAT

```

_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program-6$ docker tag program-6 a2003nu/program-6:latest
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program-6$ docker login -u a2003nu

Info → A Personal Access Token (PAT) can be used instead.
        To create a PAT, visit https://app.docker.com/settings

Password:
Login Succeeded

```

5. Push the image to the docker hub

```

_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program-6$ docker push a2003nu/program-6:latest
The push refers to repository [docker.io/a2003nu/program-6]
1351408a757a: Pushed
a1042c1cc20f: Pushed
2a7d948f67ea: Pushed
6f3502ca8022: Pushed
82140d9a70a7: Mounted from library/node
f3b40b0cdb1c: Mounted from library/node
0b1f26057bd0: Mounted from library/node
08000c18d16d: Mounted from library/node
latest: digest: sha256:fa813b14288cb59a8651b489205b19fb90812f25106a2c212be432073169ef9b size: 1993

```

6. Tag the image for the github and login to the github container registry using PAT

```

_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program-6$ docker tag program-6 ghcr.io/a2003nu/program-6:latest
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program-6$ docker login ghcr.io -u a2003nu
Password:

WARNING! Your credentials are stored unencrypted in '/home/_1RV24MC019_anusha_s/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded

```

7. Push the image to the github

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program-6$ docker push ghcr.io/a2003nu/program-6:latest
The push refers to repository [ghcr.io/a2003nu/program-6]
1351408a757a: Pushed
a1042c1cc20f: Pushed
2a7d948f67ea: Pushed
6f3502ca8022: Mounted from a2003nu/pro
82140d9a70a7: Mounted from a2003nu/pro
f3b40b0cdb1c: Mounted from a2003nu/pro
0b1f26057bd0: Mounted from a2003nu/pro
08000c18d16d: Mounted from a2003nu/pro
latest: digest: sha256:fa813b14288cb59a8651b489205b19fb90812f25106a2c212be432073169ef9b size: 1993
```

8. Add the local project to the github:

- Add the remote repository,(if not already linked)

git remote add origin <https://github.com/a2003nu/program-6.git>

- If the remote repository already exists(but incorrect),

git remote set-url origin <https://github.com/a2003nu/program-6.git>

9. Add the source code to the remote repository

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program-6$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/_1RV24MC019_anusha_s/program-6/.git/
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program-6$ git add .
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program-6$ git commit -m "Initial co
[master (root-commit) 7618d1b] Initial commit
589 files changed, 65682 insertions(+)
create mode 100644 Dockerfile
create mode 100644 file.txt
create mode 100644 node_modules/.package-lock.json
create mode 100644 node_modules/accepts/HISTORY.md
create mode 100644 node_modules/accepts/LICENSE
create mode 100644 node_modules/accepts/README.md
create mode 100644 node_modules/accepts/index.js
```

```

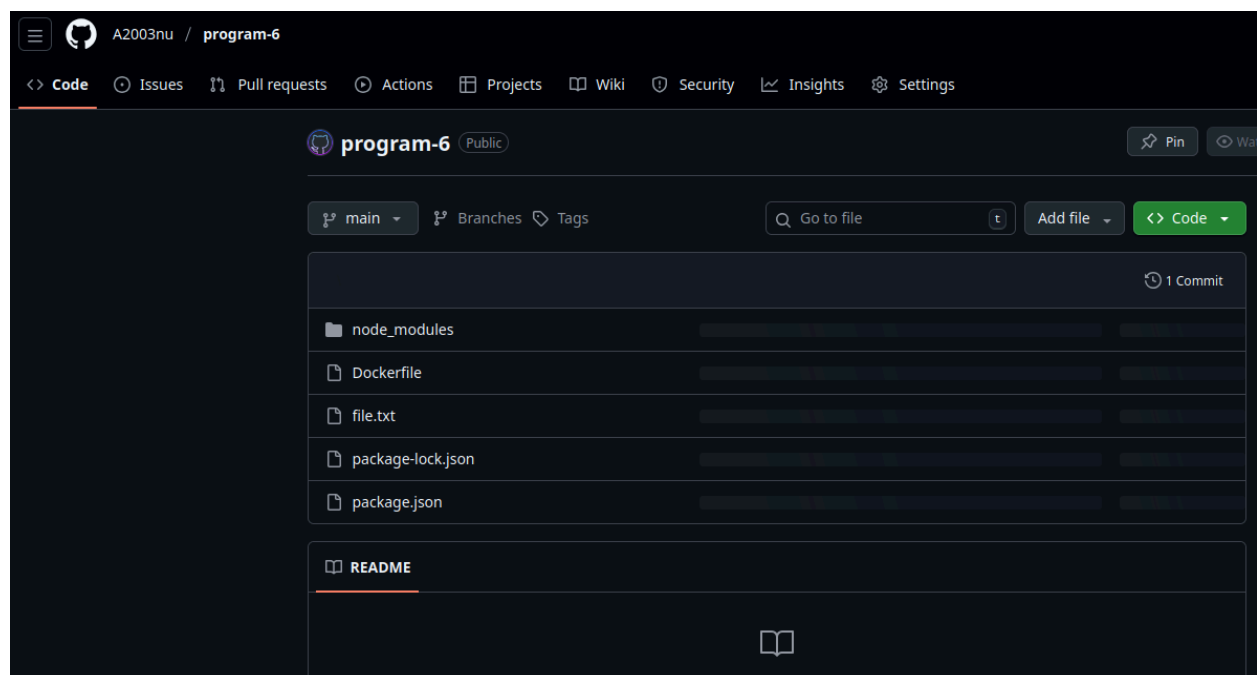
1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program-6$ git branch -M main
1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program-6$ git remote add origin https://github.com/A2003nu/program-6.git
1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program-6$ git push -u origin main
Username for 'https://github.com': a2003nu
Password for 'https://a2003nu@github.com':
Enumerating objects: 677, done.
Counting objects: 100% (677/677), done.
Delta compression using up to 8 threads
Compressing objects: 100% (625/625), done.
Writing objects: 100% (677/677), 698.32 KiB | 3.58 MiB/s, done.
Total 677 (delta 117), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (117/117), done.
To https://github.com/A2003nu/program-6.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

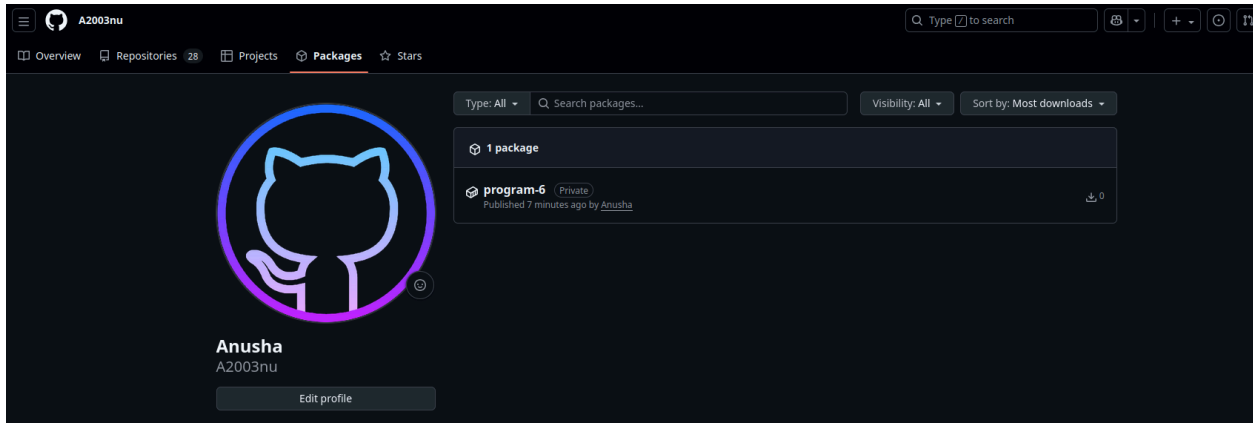
```

10. Check the uploads in the both registries

In the github:

- Check whether the source code is present in the repository
- Under your profile in the packages the particular image is present.





In the dockerhub:

- Under your repository check whether the particular image a2003nu/program-6 is visible

